

MATH ASSIGNMENT.

AKELLO ESTHER OKALANY

25/U/BIO/023/GV

I wrote C program using DevC/C++ software for the assignment.

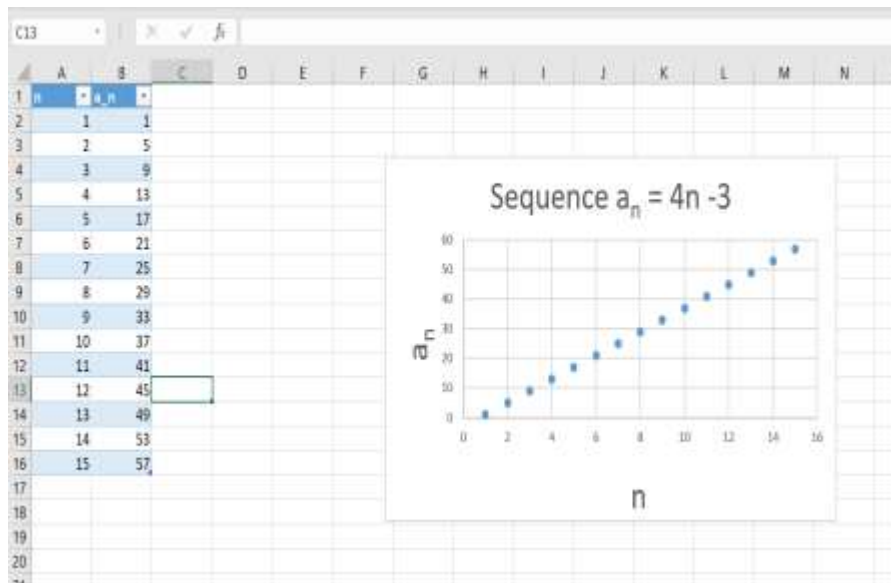
```
[*] mt.c
1  #include <stdio.h>
2  int main() {
3      int n, a, sum = 0; // declare variables
4      // n = term number, a = value of term, sum = total sum
5      // sum = 0 meaning starting from 0
6      printf("n\t a_n\n"); // print table headers for n and a_n
7      printf("-----\n");
8
9      //Loop through n = 1 to 15
10     for(n = 1; n <= 15; n++) {
11         a = 4*n - 3; // calculate the nth term using formula
12
13         printf("%d\t %d\n", n, a); // print term number and value in table format
14         sum += a; // add this term to the running sum
15     }
16
17     // after loop ends, print the total sum
18     printf("-----\n");
19     printf("Sum of first 15 terms = %d\n", sum);
20
21     return 0;
22 }
23
```

For part a) This is the output of the C program showing the first 15 terms of the sequence $a_n = 4n - 3$.

c) The program also computed the sum of the first 15 terms as shown;

```
C:\Users\User\Desktop\AKELLO-C\mtc.exe
n      a_n
-----
1       1
2       5
3       9
4      13
5      17
6      21
7      25
8      29
9      33
10     37
11     41
12     45
13     49
14     53
15     57
-----
Sum of first 15 terms = 435
```

b) The graph and the table, both using Excel.



d) The explanation;

The sequence $a_n = 4n - 3$ is an arithmetic progression with first term $a_1 = 1$ and common difference, $d = 4$ from $a_{n+1} - a_n$ for any 2 consecutive numbers in the sequence.

Each term increases by 4 thus growth is steady, forming a straight-line pattern when plotted.

The sum of the first 15 terms is **465**, which can also be verified using the formula $S_n = n/2 \times (\text{first term} + \text{last term}) = 15/2 \times (1 + 57) = 465$.